

IAL Physics Unit 1 (WPH11): Rapid Revision

Last-minute summary · Mechanics and Materials · 1 h 30 min · 80 marks · 10 MCQ + structured + one 6-mark answer

EXAM RULES THAT COST MARKS

- “Show that”: give 1 more s.f. than printed; a reverse calculation loses the final mark. If stuck, use the printed value next.
- Missing or wrong unit loses the final mark. $g = 9.81 \text{ N kg}^{-1}$ (9.8 accepted; **10 loses a mark**).
- “Is it correct?”: your value, then a numerical comparison ($2.1 \text{ m} < 3.0 \text{ m}$), then a conclusion. ecf applies.
- Scale drawing asked: trig earns only the final mark. Arrows 6–8 cm, scale stated, nose-to-tail.
- Free-body diagrams: –1 per extra force; arrows touch the dot; named forces; balanced forces drawn equal.
- Refused: “equal and opposite reaction” for N3; “human error” (say reaction time → random error).
- 6-marker: 6 linked points for 6/6. Write 6–7 numbered cause → effect sentences; name the law used.

UNITS AND VECTORS

- Vectors:** displacement, velocity, acceleration, force (W , U , drag), momentum, moment. **Scalars:** distance, speed, mass, time, energy, work, power, density, pressure, stress, strain, E , viscosity.
- Base units:** $N = \text{kg m s}^{-2}$; $J = \text{kg m}^2 \text{ s}^{-2}$; $W = \text{kg m}^2 \text{ s}^{-3}$; $\text{Pa} = \text{kg m}^{-1} \text{ s}^{-2}$; momentum $\text{kg m s}^{-1} = \text{N s}$; viscosity Pa s .
- Convert $\text{mm} \rightarrow \text{m}$ **before** squaring/cubing: $1 \text{ mm}^2 = 10^{-6} \text{ m}^2$. $\rho_{\text{water}} 1000$, seawater $\approx 1030 \text{ kg m}^{-3}$; $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$.
- Resolve: $F \cos \theta$ (along θ), $F \sin \theta$ (across). Slope: $mg \sin \theta$ along, $mg \cos \theta$ perpendicular.

KINEMATICS

$v = u + at$; $s = \frac{1}{2}(u + v)t$	constant acceleration only
$s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$	up = +, so $a = -9.81$
s-t graph	gradient = velocity (tangent)
v-t graph	gradient = a; area = displacement
a-t graph	area = change in velocity

MEASURING G

Light gate: $v = \text{card length} \div \text{blocking time}$; vary drop height s ; plot v^2 against s : **gradient = $2g$** .
From rest: $s = \frac{1}{2}gt^2$, plot s against t^2 : **gradient = $g/2$** .
Stopwatch → reaction time → large random error. Curved slide: a not constant, so use energy, not suvat.

PROJECTILE ROUTINE (ALMOST EVERY PAPER)

- Resolve: $u_H = u \cos \theta$, $u_V = u \sin \theta$.
- Horizontal $a = 0$ ($s = u_H t$); vertical $a = -9.81$.
- Time links the two directions:** find t in one, use it in the other.
- Max height: $v_V = 0$, $s = u_V^2/2g$ (+ launch height).
- Compare with the target and conclude.

- Horizontal launch: fall time depends only on height. Landing v : Pythagoras; angle = $\tan^{-1}(v_V/v_H)$.
- Air resistance:** v_H decreases; greater deceleration going up → lower max height, shorter time, shorter range, not symmetrical.

FORCES AND NEWTON’S LAWS

N1	rest or constant velocity unless a resultant force acts
N2: $\Sigma F = ma$	find the resultant first
N3 pair	equal size, opposite direction, same type, different bodies , same time

- Weight and normal contact force on one object are **never** an N3 pair. Partner of “Earth pulls you” = “you pull the Earth”.
- Lift:** $R - mg = ma$. Scales read R : $> mg$ accelerating up; $< mg$ accelerating down or slowing while rising.
- Rocket:** $a = (\text{thrust} - mg)/m$; a rises as fuel burns (m falls).
- Slope:** frictionless $a = g \sin \theta$ (mass-independent); constant speed → $mg \sin \theta = \text{resistive force}$.

MOMENTUM

CONSERVATION OF MOMENTUM (2 MARKS)

Total momentum before = total momentum after (1), **provided no external force acts / closed system** (1). The second mark is most often lost.

- Choose + direction; negative answer = moves the other way (say so).
- Explosion/recoil: initial $p = 0$, so equal and opposite momenta; heavier part slower.
- KE usually not conserved: work done deforming → thermal energy. KE lost = $\Sigma \frac{1}{2}mv^2$ before – after.
- $F = \Delta p/\Delta t$; jet: force = (mass per second) $\times v$.
- Link to N3:** equal and opposite forces for the same time → equal and opposite Δp → total conserved; lighter object has larger Δv .

MOMENTS AND EQUILIBRIUM

Moment	$F \times \text{perpendicular distance from pivot to line of action}$ ($F \sin \theta \times d$)
Principle of moments	in equilibrium, $\Sigma \text{ clockwise} = \Sigma \text{ anticlockwise}$ about any point
Equilibrium	zero resultant force and zero resultant moment
Centre of gravity	point where the weight may be taken to act
Toppling	line of action of W passes outside the base → unbalanced moment

TWO UNKNOWN SUPPORTS

- Take moments about one support (its force has zero moment) → find the other.
- Then total up = total down. Hinge: find F_H , F_V , then Pythagoras + tan.

- Load moves away from $X \rightarrow$ its moment about X rises → the far support force rises (total unchanged).
- Force at a smaller angle to the lever → smaller perpendicular component → larger force needed.

WORK, ENERGY, POWER

$\Delta W = F\Delta s \cos \theta$	Δs in the direction of F
$E_k = \frac{1}{2}mv^2$; $\Delta E_{\text{grav}} = mg\Delta h$	Δh vertical: $L \sin \theta$ on a slope
$P = \Delta W/\Delta t = Fv$	constant P : F falls as v rises
efficiency	useful output $\div$ total input

ENERGY WITH A RESISTIVE FORCE

Write the statement first: **energy at start = energy at end + work against resistive force**, e.g. $\frac{1}{2}mv^2 = mgL \sin \theta + F_R L$.
Frictionless: $v = \sqrt{2g\Delta h}$, independent of mass.
Pumping water: power = $(mg\Delta h + \frac{1}{2}mv^2)$ per second;
mass per second = ρAv .

FLUIDS

$\rho = m/V$; $V = 4/3\pi r^3$	use radius, not diameter
Upthrust = $\rho_{\text{fluid}} V_{\text{sub}} g$	= weight of fluid displaced
$F = 6\pi\eta r v$ (Stokes)	small, spherical, low speed, laminar
Falling: $W = U + D$	rising bubble: $U = W + D$

- **Laminar:** layers don't mix; velocity at a point constant.
- **Turbulent:** layers mix, eddies, velocity changes.
- Hotter liquid $\rightarrow$ lower $\eta \rightarrow$ less drag $\rightarrow$ higher terminal velocity.
- Floating: $U = W$. Denser (sea) water $\rightarrow$ less volume displaced $\rightarrow$ floats higher.
- Terminal v rises with size: W , $U \propto r^3$ but $D \propto r$.

STOKES' LAW CALCULATION

1. $V = 4/3\pi r^3$; $W = \rho_s Vg$, $U = \rho_f Vg$; $D = W - U = (\rho_s - \rho_f)Vg$.
2. $\eta = D/(6\pi r v)$; v = distance $\div$ time between markers.
3. "Does Stokes apply?" Compare $6\pi\eta r v$ with $W - U$; big mismatch $\rightarrow$ not laminar.

- **Viscometer:** first marker low enough for terminal v (equal times for equal sections); repeat; constant temperature.

MATERIALS

$\Delta F = k\Delta x$	k = gradient of F - x (to limit of proportionality)
$E_{\text{el}} = \frac{1}{2}F\Delta x = \frac{1}{2}k\Delta x^2$	area under F - x ; trapezium between loads
$\sigma = F/A$; $\epsilon = \Delta x/x$; $E = \sigma/\epsilon$	E = gradient $\times x/A$
area under σ - ϵ	energy per unit volume

- Series springs: extension adds, k halves. Parallel: extension halves, k doubles.
- σ - ϵ order: limit of proportionality $\rightarrow$ elastic limit $\rightarrow$ yield point $\rightarrow$ UTS $\rightarrow$ breaking point.
- **Stiff** large E $\cdot$ **strong** high breaking stress $\cdot$ **ductile** large plastic region $\cdot$ **brittle** little plastic deformation $\cdot$ **tough** large area under graph $\cdot$ **hard** resists scratching.
- Elastic: returns to original length. Plastic: permanent. Loading-unloading loop area = energy to thermal.

YOUNG MODULUS EXPERIMENT

1. Long, thin wire + initial load to keep it taut; marker and metre rule (or vernier).
2. Measure original length x from clamp to marker.
3. Micrometer: diameter at several points and orientations; mean; $A = \pi d^2/4$.
4. Add masses; $F = mg$; extension = new - original reading, eye level (parallax).
5. Plot stress against strain; E = gradient of the straight section.
6. Goggles in case the wire snaps. Long thin wire $\rightarrow$ large extension $\rightarrow$ small % uncertainty.

THE 6-MARKERS TO MEMORISE

A • Skydiver to terminal velocity

1. $v = 0$, no drag, $a = g$
2. v rises, so drag rises
3. resultant ($W - D$) falls, so a falls
4. $D = W$: resultant zero, terminal velocity
5. parachute opens: $D > W$, resultant up, decelerates
6. D falls with speed until $D = W$: new, lower terminal velocity

B • Balloon or bubble rising

1. $U > W$ at release (no drag)
2. resultant up, so accelerates (N_2)
3. v rises, so drag (down) rises; U , W constant
4. resultant falls, so a falls
5. $U = W + D$: resultant zero
6. constant (terminal) velocity

C • Bungee jumper

1. rope stretching: $W > T$, resultant down
2. acceleration down
3. T rises with extension (Hooke)
4. resultant and a decrease
5. $T = W$: $a = 0$, maximum speed
6. $T > W$: resultant up, decelerates to rest

D • N2 + N3 interaction (ball hits skittle)

1. ball exerts a force on skittle
2. skittle exerts a force on ball
3. N_3 : equal size, opposite direction
4. resultant force on each object
5. N_2 : each accelerates in direction of its resultant
6. skittle accelerates forward; ball decelerates

E • Scales in a lift

1. reading = normal contact force on student
2. constant speed: resultant zero, so $R = W$
3. reading = weight
4. decelerating upward: $R < W$
5. resultant force downward
6. reading less than weight

F • Motion from an s-t graph

1. gradient = velocity
2. constant gradient $\rightarrow$ constant v
3. gradient falling $\rightarrow v$ falling
4. so decelerating
5. gradient zero $\rightarrow$ momentarily at rest
6. negative gradient $\rightarrow$ moving back

MCQ TRAPS

- Circular motion at constant speed: p changes, KE constant.
- Area under σ - ϵ = work per unit **volume**.
- Gradient of s vs t^2 is $g/2$; of v^2 vs s is $2g$.
- Doubling diameter needs $4\times$ tension for the same stress. Submerged sinking object: U constant, only drag changes.
- Breaking stress, ρ , E are material properties; breaking force is not.